

Storage Classes

Storage Classes in C decide the part of storage to allocate memory for a variable. It also determines the scope of a variable defined in a C program gets some physical location in memory where the variable value is stored.

Types of Storage Class (Specifier)

There are 4 storage class specifier in C language.

1) Auto 2) Register 3) Extern 4) Static.

Auto : (Automatic) - A variable defined within a function or block with auto keyword belongs to automatic storage class. Variables having automatic storage class, are local to the block which they are defined and get destroyed on exit from the block.

Example :

```
#include <stdio.h>
#include <conio.h>

void main()
{
    auto int a=1;
    {
        printf("%d", a);
    }
    auto int a=2;
    {
        auto int a=3;
        printf("%d", a);
    }
    printf("%d", a);
}
```


Register

Register: Register specifier declares a variable of register storage class. Variables belong to register storage class are local to the block which they are defined in and get destroyed on exit from the block.

Example: #include <stdio.h>
#include <conio.h>

void main()

{ register int x = 10, *P;

P = &x;

printf("Value = %d", *P);

printf("Address = \n", P);

void add()

{ printf("%d", *P); }

}

}

↓
This variable will not work.

Static: If we declared a variable as static then that variable value can't change within the scope of the variable, but the variable value can change outside the scope of the variable declaration.

Example :

```
#include <stdio.h>
#include <conio.h>
void static Demo (
{
    static int i;
    {
        static int i = 1;
        printf("%d", i);
        i++;
    }
    printf("%d\n", i);
    i++;
}
void main()
{
    static Demo();
    static Demo();
}
```

1 = 2

Extern : External storage class is used to specify that a variable is declared with external linkage else-where in the program we have to assume that variable has to be assigned anywhere within the program.

Example :

```
#include <stdio.h>
#include <conio.h>
extern int x;
void main()
{
    printf("%d\n", x);
}
int x = 10;
```